

# Probability Preliminaries

The purpose of these notes is to give a self-contained introduction to measure theoretic probability that would be useful for the remainder of the course[cite: 5]. The learning objective here for the purposes of the course is to familiarise oneself with the language of the probability in measure theoretic terms so that one can follow the concepts that build upon it[cite: 6].

## 1 $\sigma$ -algebra

**Definition 1.1.** Let  $\Omega$  be a non-empty set, and let  $\mathcal{F}$  be a collection of subsets of  $\Omega$ [cite: 8]. We say that  $\mathcal{F}$  is a  $\sigma$ -algebra (or a  $\sigma$ -field) if[cite: 8]:

1. **(Null)**  $\emptyset \in \mathcal{F}$  [cite: 9]
2. **(Complement)** If  $A \in \mathcal{F}$  then  $A^c \in \mathcal{F}$ , where  $A^c$  is the complement of  $A$ [cite: 10].
3. **(Countable union)** If  $A_1, A_2, \dots \in \mathcal{F}$  then  $\cup_{n=1}^{\infty} A_n \in \mathcal{F}$ [cite: 11].

*Remark.* A  $\sigma$ -algebra represents the collection of events whose occurrence can be meaningfully observed[cite: 12]. Different  $\sigma$ -algebras correspond to different levels of information available to an observer[cite: 13].

**Example 1.1.** Consider tossing a fair coin twice[cite: 14]. The sample space is[cite: 14]:

$$\Omega = \{HH, HT, TH, TT\}$$

A simple example of a  $\sigma$ -algebra on  $\Omega$  is[cite: 16]:

$$\mathcal{F} = \{\emptyset, \Omega\}$$

This is called the trivial  $\sigma$ -algebra[cite: 18]. A less trivial  $\sigma$ -algebra is[cite: 18]:

$$\mathcal{F} = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}$$

which corresponds to observing only the outcome of the first toss[cite: 20].

*Remark.* The restriction to  $\sigma$ -algebras is not merely a matter of convenience[cite: 21]. In general, it is impossible to assign probabilities consistently to all subsets of a sample space[cite: 22]. A  $\sigma$ -algebra specifies a collection of events on which a probability measure can be defined in a mathematically consistent way[cite: 23].

The elements of  $\mathcal{F}$  are subsets of  $\Omega$  and are called  $\mathcal{F}$ -events or  $\mathcal{F}$ -measurable sets[cite: 31]. If there's no ambiguity we simply say event or measurable set[cite: 31, 32]. The pair  $(\Omega, \mathcal{F})$  is said to be a measurable space[cite: 32].

**Definition 1.2.** If  $\mathcal{A}$  is a collection of subsets of  $\Omega$ , then there is a smallest  $\sigma$ -field on  $\Omega$  that contains  $\mathcal{A}$ , which is contained in every  $\sigma$ -field that contains  $\mathcal{A}$ [cite: 38]. We denote this  $\sigma$ -field by  $\mathcal{F}(\mathcal{A})$  and refer to it as the  $\sigma$ -field generated by  $\mathcal{A}$ [cite: 38].

*Remark.* It is common to write  $\sigma(\mathcal{A})$  for the  $\sigma$ -field generated by  $\mathcal{A}$ [cite: 39]. Thus  $\mathcal{F}(\mathcal{A}) = \sigma(\mathcal{A})$ [cite: 39].

**Example 1.2.** Let  $\Omega = \{1, 2, 3, 4\}$  and let  $\mathcal{A} = \{\{1, 2\}\}$ [cite: 40]. Then the smallest  $\sigma$ -field containing  $\mathcal{A}$  must contain  $\emptyset$  and  $\Omega$ , and it must be closed under complements[cite: 40]. Hence it must contain  $\{1, 2\}^c = \{3, 4\}$ [cite: 41, 43]. Therefore[cite: 42]:

$$\mathcal{F}(\mathcal{A}) = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$$

**Example 1.3.** Let  $\Omega = \{HH, HT, TH, TT\}$  and let  $\mathcal{A} = \{\{HH, HT\}\}$ [cite: 45]. The event  $\{HH, HT\}$  is “the first toss is Heads”[cite: 45]. Its complement is  $\{TH, TT\}$ [cite: 46]. Hence[cite: 46]:

$$\mathcal{F}(\mathcal{A}) = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}$$

This is precisely the  $\sigma$ -field corresponding to observing only the first toss[cite: 48].

## 1.1 Filtration

**Definition 1.3.** Let  $(\Omega, \mathcal{F})$  be a measurable space[cite: 57].

- (a) A discrete filtration on  $(\Omega, \mathcal{F})$  is an increasing sequence of  $\sigma$ -algebras  $(\mathcal{F}_n)_{n \geq 0}$  such that  $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}_n \subset \cdots \subset \mathcal{F}$ [cite: 58, 59].
- (b) A continuous filtration on  $(\Omega, \mathcal{F})$  is a family of  $\sigma$ -algebras  $(\mathcal{F}_t)_{t \in I}$ , where  $I \subset \mathbb{R}$  is an interval, such that for all  $s, t \in I$  with  $s < t$ ,  $\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ [cite: 60, 61, 62, 63].

The  $\sigma$ -algebra  $\mathcal{F}_n$  (respectively  $\mathcal{F}_t$ ) represents the information available up to time  $n$  (respectively time  $t$ ) [cite: 64].

**Example 1.4.** Consider repeatedly tossing a fair coin[cite: 65]. The sample space is[cite: 65]:

$$\Omega = \{H, T\}^{\mathbb{N}} = \{\omega = (\omega_1, \omega_2, \dots) : \omega_i \in \{H, T\}\}$$

Define the coordinate value  $X_n(\omega) = \omega_n$ ,  $n \geq 1$  so that  $X_n$  records the outcome of the  $n$ -th toss[cite: 67, 68, 69, 70].

For each  $n \geq 1$ , let  $\mathcal{F}_n$  be the  $\sigma$ -field generated by the first  $n$  tosses[cite: 71]:

$$\mathcal{F}_n = \sigma(X_1, \dots, X_n)$$

Equivalently,  $\mathcal{F}_n$  consists of all events that can be decided by knowing only  $(X_1, \dots, X_n)$ [cite: 73]. The first toss can be either  $H$  or  $T$ , so there are only two basic events[cite: 73]:

$$\{X_1 = H\} = \{\omega : \omega_1 = H\}$$

$$\{X_1 = T\} = \{\omega : \omega_1 = T\}$$

Thus[cite: 74],

$$\mathcal{F}_1 = \{\emptyset, \Omega, \{X_1 = H\}, \{X_1 = T\}\}$$

In words, at time 1 you only know whether the first toss was  $H$  or  $T$ [cite: 78]. After two tosses, the basic events are[cite: 78]:

$$\{X_1 = H, X_2 = H\}, \quad \{X_1 = H, X_2 = T\}, \quad \{X_1 = T, X_2 = H\}, \quad \{X_1 = T, X_2 = T\}$$

Therefore  $\mathcal{F}_2$  consists of all unions of these four sets (and also  $\emptyset$  and  $\Omega$ ) [cite: 80]. For example[cite: 80]:

$$\{X_2 = H\} = \{X_1 = H, X_2 = H\} \cup \{X_1 = T, X_2 = H\} \in \mathcal{F}_2$$

$$\{X_1 = X_2\} = \{X_1 = H, X_2 = H\} \cup \{X_1 = T, X_2 = T\} \in \mathcal{F}_2$$

But the event  $\{X_2 = H\}$  is not in  $\mathcal{F}_1$  since it cannot be decided from the first toss alone[cite: 84]. Knowing the first two tosses certainly tells you the first toss[cite: 84, 85]. Formally[cite: 85]:

$$\{X_1 = H\} = \{X_1 = H, X_2 = H\} \cup \{X_1 = H, X_2 = T\} \in \mathcal{F}_2$$

and similarly  $\{X_1 = T\} \in \mathcal{F}_2$ [cite: 87]. Hence every event in  $\mathcal{F}_1$  is also in  $\mathcal{F}_2$ , so  $\mathcal{F}_1 \subset \mathcal{F}_2$ [cite: 87]. Continuing in this way yields an increasing sequence which is a discrete filtration[cite: 88, 89]:

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{F}_n \subset \cdots$$

## 2 Probability measure

**Definition 2.1.** Let  $\Omega$  be a nonempty set, and let  $\mathcal{F}$  be a  $\sigma$ -algebra of subsets of  $\Omega$ [cite: 93]. A probability measure  $\mathbb{P}$  is a function that, to every set  $A \in \mathcal{F}$ , assigns a number in  $[0, 1]$ , called the probability of  $A$  and written  $\mathbb{P}(A)$ [cite: 93].

1.  $\mathbb{P}(\Omega) = 1$  [cite: 94]
2. If  $A_1, A_2, \dots$  is a sequence of disjoint sets in  $\mathcal{F}$ , then[cite: 95]:

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$$

The triple  $(\Omega, \mathcal{F}, \mathbb{P})$  is called a probability space[cite: 97].

**Example 2.1.** Consider tossing a fair coin twice, with sample space  $\Omega = \{HH, HT, TH, TT\}$ [cite: 98]. Let  $\mathcal{F} = 2^\Omega$  be the power set of  $\Omega$  (i.e., all subsets of  $\Omega$ )[cite: 99]. Define  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  by assigning equal probability to each outcome[cite: 100]:

$$\mathbb{P}(\{HH\}) = \mathbb{P}(\{HT\}) = \mathbb{P}(\{TH\}) = \mathbb{P}(\{TT\}) = \frac{1}{4}$$

Then  $(\Omega, \mathcal{F}, \mathbb{P})$  is a probability space[cite: 102]. For instance[cite: 102]:

$$\mathbb{P}(\text{exactly one Head}) = \mathbb{P}(\{HT, TH\}) = \frac{1}{2}$$

**Example 2.2.** Consider again  $\Omega = \{HH, HT, TH, TT\}$ , but now take[cite: 106]:

$$\mathcal{F} = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}$$

which corresponds to observing only the first toss[cite: 108]. Define  $\mathbb{P}$  on  $\mathcal{F}$  by[cite: 108]:

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\Omega) = 1$$

$$\mathbb{P}(\{HH, HT\}) = \frac{1}{2}$$

$$\mathbb{P}(\{TH, TT\}) = \frac{1}{2}$$

Then  $\mathbb{P}$  is a probability measure on  $(\Omega, \mathcal{F})$ [cite: 112]. Note that events such as  $\{HH\}$  are not in  $\mathcal{F}$ , so under this information structure one cannot even ask for  $\mathbb{P}(\{HH\})$ [cite: 113].

## 2.1 Borel $\sigma$ -field

**Definition 2.2.** The Borel  $\sigma$ -field on  $\mathbb{R}$ , denoted by  $\mathcal{B}(\mathbb{R})$ , is the  $\sigma$ -field generated by the open intervals in  $\mathbb{R}$ [cite: 115]. That is[cite: 115]:

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b, a, b \in \mathbb{R}\})$$

The elements of  $\mathcal{B}(\mathbb{R})$  are called Borel sets[cite: 117].

*Remark.* The Borel  $\sigma$ -field can equivalently be generated by closed intervals[cite: 118]. In particular[cite: 118]:

$$\mathcal{B}(\mathbb{R}) = \sigma(\{[a, b] : a \leq b, a, b \in \mathbb{R}\})$$

More generally, the same  $\sigma$ -field is obtained if we start with open intervals, closed intervals, half-open intervals, or rays of the form  $(-\infty, a]$  or  $(a, \infty)$ [cite: 120].

## 2.2 Random variable

**Definition 2.3.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space[cite: 123]. A random variable is a function [cite: 123]

$$X : \Omega \rightarrow \mathbb{R}$$

such that for every Borel set  $B \in \mathcal{B}(\mathbb{R})$ [cite: 125],

$$\{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$$

In this case,  $X$  is said to be  $\mathcal{F}$ -measurable[cite: 127].

*Remark.* The measurability condition is equivalent to requiring that  $\{\omega \in \Omega : X(\omega) \leq a\} \in \mathcal{F}$  for all  $a \in \mathbb{R}$ [cite: 128, 129].

*Remark.* The measurability requirement ensures that events of the form  $\{X \in B\}$  are events in  $\mathcal{F}$  and therefore have well-defined probabilities[cite: 130, 131, 132]. Without this requirement, it would be possible to define functions whose value depends on the outcome, but for which probabilities such as  $\mathbb{P}(X \leq a)$  are not defined[cite: 132].

**Example 2.3.** Consider tossing a fair coin once[cite: 133]. The sample space is  $\Omega = \{H, T\}$ , with  $\mathcal{F} = 2^\Omega$  and  $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = \frac{1}{2}$ [cite: 133, 134, 135]. Define a function  $X : \Omega \rightarrow \mathbb{R}$  by[cite: 136]:

$$X(H) = 1, \quad X(T) = 0$$

$X$  assigns a numerical value to each outcome[cite: 138]. Let  $B = (-\infty, 0.5]$ [cite: 139]. Then[cite: 139]:

$$\{\omega \in \Omega : X(\omega) \in B\} = \{T\} \in \mathcal{F}$$

Similarly[cite: 140],

$$\{\omega : X(\omega) > 0.5\} = \{H\} \in \mathcal{F}$$

Since the preimage of every Borel set is an event in  $\mathcal{F}$ , the function  $X$  is a random variable[cite: 143].

## 2.3 Distribution of a random variable

**Definition 2.4.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and  $X : \Omega \rightarrow \mathbb{R}$  a random variable[cite: 145]. The distribution (or law) of  $X$  is the probability measure  $\mathbb{P}_X$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  defined by[cite: 146]:

$$\mathbb{P}_X(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{B}(\mathbb{R})$$

**Definition 2.5.** The cumulative distribution function (CDF) of  $X$  is [cite: 148]

$$F_X(x) := \mathbb{P}(X \leq x) = \mathbb{P}_X((-\infty, x]), \quad x \in \mathbb{R}$$

*Remark.* Let  $\lambda$  denote the Lebesgue measure on  $\mathbb{R}$ [cite: 150]. A probability measure  $\mu$  on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$  is said to have a density  $f$  (with respect to Lebesgue measure) if[cite: 150, 151]:

$$\mu(B) = \int_B f(x)\lambda(dx) = \int_B f(x)dx, \quad B \in \mathcal{B}(\mathbb{R})$$

where  $f \geq 0$  and  $\int_{\mathbb{R}} f(x)dx = 1$ [cite: 154]. In this case we write informally  $\mu(dx) = f(x)dx$ [cite: 154].

**Example 2.4.** Let  $\Omega = (0, 1)$ ,  $\mathcal{F} = \mathcal{B}((0, 1))$ , and define  $\mathbb{P}$  by  $\mathbb{P}(A) = \lambda(A)$  for  $A \in \mathcal{B}((0, 1))$  where  $\lambda$  is Lebesgue measure restricted to  $(0, 1)$ [cite: 156, 157, 158, 159]. Then  $(\Omega, \mathcal{F}, \mathbb{P})$  is a probability space, and the identity map  $X(\omega) = \omega$  is a random variable[cite: 159, 160, 161]. For any Borel set  $B \in \mathcal{B}(\mathbb{R})$ [cite: 161],

$$\mathbb{P}_X(B) = \mathbb{P}(X \in B) = \mathbb{P}((0, 1) \cap B) = \lambda((0, 1) \cap B)$$

Equivalently,  $\mathbb{P}_X$  has density  $f(x) = 1_{(0,1)}(x)$  with respect to Lebesgue measure, i.e.,[cite: 163, 164, 165]:

$$\mathbb{P}_X(B) = \int_B 1_{(0,1)}(x)dx$$

This is the uniform distribution on  $(0, 1)$ , often written  $X \sim \text{Unif}(0, 1)$ [cite: 167].

**Example 2.5.** A random variable  $X$  is said to have a normal distribution with mean  $\mu$  and variance  $\sigma^2 > 0$  (written  $X \sim N(\mu, \sigma^2)$ ) if its law  $\mathbb{P}_X$  is absolutely continuous with respect to Lebesgue measure and has density[cite: 168]:

$$\varphi_{\mu, \sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

In measure form, for every Borel set  $B \in \mathcal{B}(\mathbb{R})$ [cite: 170],

$$\mathbb{P}_X(B) = \int_B \varphi_{\mu, \sigma}(x)dx$$

In particular, the CDF is[cite: 171]:

$$F_X(x) = \mathbb{P}_X((-\infty, x]) = \int_{-\infty}^x \varphi_{\mu, \sigma}(t)dt$$

*Remark.* In both examples, the distribution is a measure on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ [cite: 174]. The density notation  $f(x)dx$  is simply a convenient way to describe the measure  $\mathbb{P}_X$  in terms of Lebesgue measure[cite: 174].

## 2.4 $\sigma$ -sigma-field generated by a random variable

**Definition 2.6.** Let  $(\Omega, \mathcal{F})$  be a measurable space and let  $X : \Omega \rightarrow \mathbb{R}$  be a random variable[cite: 176]. The  $\sigma$ -field generated by  $X$  is[cite: 177]:

$$\sigma(X) := \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

More generally, for random variables  $X_1, \dots, X_n$ [cite: 179]:

$$\sigma(X_1, \dots, X_n) := \sigma(\sigma(X_1) \cup \dots \cup \sigma(X_n))$$

**Example 2.6.** Consider tossing a fair coin once:  $\Omega = \{H, T\}$ ,  $\mathcal{F} = 2^\Omega$ [cite: 181, 182]. Define the random variable  $X : \Omega \rightarrow \mathbb{R}$  by  $X(H) = 1$  and  $X(T) = 0$ [cite: 183, 184, 186]. Then  $X(\Omega) = \{0, 1\}$ [cite: 187]. By definition[cite: 187],

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

Since  $X$  only takes the two values  $\{0, 1\}$ , the preimage  $X^{-1}(B)$  depends only on which of 0 and 1 lie in  $B$ [cite: 189]. More precisely, for any Borel set  $B \subseteq \mathbb{R}$ [cite: 189],

$$X^{-1}(B) = \begin{cases} \emptyset, & \text{if } B \cap \{0, 1\} = \emptyset, \\ \{T\}, & \text{if } B \cap \{0, 1\} = \{0\}, \\ \{H\}, & \text{if } B \cap \{0, 1\} = \{1\}, \\ \Omega, & \text{if } B \cap \{0, 1\} = \{0, 1\}. \end{cases}$$

Each of these four cases occurs for some Borel set  $B$  (e.g.  $B = \emptyset, \{0\}, \{1\}, \{0, 1\}$ ), so[cite: 191]:

$$\sigma(X) = \{\emptyset, \{H\}, \{T\}, \Omega\} = 2^\Omega = \mathcal{F}$$

In this one-toss example, observing  $X$  tells you exactly whether the outcome was  $H$  or  $T$ , so the information in  $\sigma(X)$  is the same as the full information  $\mathcal{F}$ [cite: 193].

**Example 2.7.** Consider tossing a fair coin twice:  $\Omega = \{HH, HT, TH, TT\}$ ,  $\mathcal{F} = 2^\Omega$ [cite: 194, 195, 196]. Define the random variable  $X : \Omega \rightarrow \mathbb{R}$  by letting  $X$  be the number of Heads[cite: 197]:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

Then  $X(\Omega) = \{0, 1, 2\}$ [cite: 199]. By definition[cite: 199],

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

Because  $X$  only takes the three values  $\{0, 1, 2\}$ , the preimage  $X^{-1}(B)$  depends only on which of the values 0, 1, 2 lie in  $B$ [cite: 201]. More precisely, for any Borel set  $B \subseteq \mathbb{R}$ [cite: 201],

$$X^{-1}(B) = \begin{cases} \emptyset, & \text{if } B \cap \{0, 1, 2\} = \emptyset, \\ \{TT\}, & \text{if } B \cap \{0, 1, 2\} = \{0\}, \\ \{HT, TH\}, & \text{if } B \cap \{0, 1, 2\} = \{1\}, \\ \{HH\}, & \text{if } B \cap \{0, 1, 2\} = \{2\}, \\ \{TT, HT, TH\}, & \text{if } B \cap \{0, 1, 2\} = \{0, 1\}, \\ \{TT, HH\}, & \text{if } B \cap \{0, 1, 2\} = \{0, 2\}, \\ \{HT, TH, HH\}, & \text{if } B \cap \{0, 1, 2\} = \{1, 2\}, \\ \Omega, & \text{if } B \cap \{0, 1, 2\} = \{0, 1, 2\}. \end{cases}$$

Since each of the eight intersections listed above can occur for some Borel set  $B$ , the  $\sigma$ -field generated by  $X$  is exactly the collection of these eight sets[cite: 219]:

$$\sigma(X) = \{\emptyset, \{TT\}, \{HT, TH\}, \{HH\}, \{TT, HT, TH\}, \{TT, HH\}, \{HT, TH, HH\}, \Omega\}$$

In particular,  $\{HH\} = \{X = 2\} = X^{-1}(\{2\}) \in \sigma(X)$ , while  $\{HH, HT\} \notin \sigma(X)$  because it is not equal to  $X^{-1}(B)$  for any Borel  $B$  (it would have to either contain both  $HT$  and  $TH$ , or contain neither, since  $X(HT) = X(TH) = 1$ )[cite: 221].

## 2.5 Almost sure events

**Definition 2.7.** An event  $A \in \mathcal{F}$  occurs almost surely (a.s.) if  $\mathbb{P}(A) = 1$ . A statement is said to hold almost surely if it holds for all  $\omega$  outside a set of probability 0[cite: 224].

**Definition 2.8.** Two random variables  $X, Y$  are equal almost surely, written  $X = Y$  a.s., if [cite: 225]

$$\mathbb{P}(\{\omega : X(\omega) = Y(\omega)\}) = 1$$

*Remark.* In probability theory we often identify random variables that differ only on a null set, since they lead to the same probabilities and expectations[cite: 227].

**Example 2.8.** Let  $\Omega = [0, 1]$  with  $\mathcal{F} = \mathcal{B}([0, 1])$  and the uniform probability measure (i.e.  $\mathbb{P}([a, b]) = b - a$ ) [cite: 228]. Let  $A = [0, 1] \setminus \{1/2\}$  [cite: 229]. Then  $\mathbb{P}(A) = 1$  because  $\mathbb{P}(\{1/2\}) = 0$ , so  $A$  occurs almost surely [cite: 230]. However  $A$  does not occur surely (i.e. for every  $\omega$ ), because  $\omega = 1/2$  is excluded [cite: 230]. This illustrates that “almost surely” allows exceptional outcomes, as long as they have probability 0 [cite: 231].

**Example 2.9** (Equal almost surely but not equal pointwise). Let  $\Omega = [0, 1]$  with  $\mathcal{F} = \mathcal{B}([0, 1])$  and  $\mathbb{P}$  uniform [cite: 232]. Define  $X(\omega) = 0$  for all  $\omega \in [0, 1]$  and [cite: 233, 235]

$$Y(\omega) = \begin{cases} 1, & \omega = 1/2, \\ 0, & \omega \neq 1/2. \end{cases}$$

Then  $X(\omega) \neq Y(\omega)$  at  $\omega = 1/2$ , but [cite: 234]:

$$\{\omega : X(\omega) = Y(\omega)\} = [0, 1] \setminus \{1/2\} \Rightarrow \mathbb{P}(X = Y) = 1$$

Hence  $X = Y$  a.s., even though  $X \neq Y$  as functions [cite: 238].

**Example 2.10** (Why we identify a.s.-equal random variables). In the previous example,  $X$  and  $Y$  have the same distribution: for any Borel set  $B \subseteq \mathbb{R}$  [cite: 239],

$$\mathbb{P}(X \in B) = \mathbb{P}(Y \in B)$$

because  $Y$  differs from  $X$  only on a probability-0 set [cite: 241]. Also [cite: 241],

$$\mathbb{E}[X] = 0 = \mathbb{E}[Y]$$

since changing values on a null set does not change the integral [cite: 243]. This is why probability theory often treats a.s.-equal random variables as “the same” [cite: 244].

## 3 Expectation

Let  $X$  be a random variable on a probability space  $(\Omega, \mathcal{F}, \mathbb{P})$  [cite: 247]. We would like to define an “average value” of  $X$  that takes the probabilities into account [cite: 248]. Then we say that the Expectation of a random variable is defined by [cite: 249]:

$$\mathbb{E}[X] = \int X d\mathbb{P}$$

if and only if  $\mathbb{E}[|X|] < \infty$  [cite: 251]. If  $\mathbb{E}[|X|] < \infty$  we say  $X$  is integrable [cite: 251].

*Remark.* In the above definition the integration is a Lebesgue integral[cite: 253]. In probability theory one works with the Lebesgue integral instead of the usual Riemann integral, because  $\Omega$  could be an “abstract” set like  $\{H, T\}$ [cite: 254]. However, the Lebesgue integration is defined only for non-negative functions[cite: 255]. This is resolved by defining  $X$  as

$$X = X^+ - X^-$$

where  $X^+ = \max(X, 0)$  and  $X^- = \max(-X, 0)$ [cite: 257].  $\mathbb{E}[X]$  exists when  $\mathbb{E}[X^+]$  and  $\mathbb{E}[X^-]$  exist[cite: 257]. The condition  $\mathbb{E}[|X|] < \infty$  ensures this[cite: 257].

Use the following definitions for calculating expectations[cite: 258]:

1. If  $X$  takes discrete values[cite: 259]:

$$\mathbb{E}[X] = \sum_i x_i \mathbb{P}(X = x_i)$$

2. Otherwise[cite: 260]:

$$\mathbb{E}[X] = \int x f(x) dx$$

where  $f(x)$  is the probability density function of  $X$  if it exists[cite: 262].

### 3.1 Properties of expectation

1. If  $X \leq Y$  a.s. and  $X, Y$  are integrable[cite: 264],

$$\mathbb{E}[X] \leq \mathbb{E}[Y]$$

2. If  $a, b \in \mathbb{R}$  and  $X, Y$  are integrable[cite: 266],

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

### 3.2 Examples

**Example 3.1.** Consider the infinite coin-toss space  $\Omega = \{H, T\}^{\mathbb{N}}$  with the product probability measure  $\mathbb{P}$  corresponding to  $\mathbb{P}(\text{Head on each toss}) = \frac{1}{2}$ [cite: 269]. Fix  $n \geq 1$  and define[cite: 270]:

$$Y_n(\omega) = \begin{cases} 1, & \omega_n = H, \\ 0, & \omega_n = T. \end{cases}$$

Then  $Y_n$  takes only the values 0 and 1, so[cite: 272]:

$$\mathbb{E}[Y_n] = 1 \cdot \mathbb{P}(Y_n = 1) + 0 \cdot \mathbb{P}(Y_n = 0) = \mathbb{P}(\omega_n = H) = \frac{1}{2}$$

**Example 3.2.** Let  $\Omega = [0, 1]$ ,  $\mathcal{F} = \mathcal{B}([0, 1])$ , and let  $\mathbb{P}$  be the uniform probability measure on  $[0, 1]$ [cite: 274]. Define[cite: 275]:

$$X(\omega) = \begin{cases} 1, & \omega \text{ is irrational,} \\ 0, & \omega \text{ is rational.} \end{cases}$$

Since  $\mathbb{Q} \cap [0, 1]$  is countable and each singleton has Lebesgue measure 0[cite: 277],

$$\mathbb{P}(\omega \text{ is rational}) = \mathbb{P}(\mathbb{Q} \cap [0, 1]) = 0, \quad \mathbb{P}(\omega \text{ is irrational}) = 1$$

Hence[cite: 278],

$$\mathbb{E}[X] = 1 \cdot \mathbb{P}(\omega \text{ is irrational}) + 0 \cdot \mathbb{P}(\omega \text{ is rational}) = 1$$



**Example 3.3** (St. Petersburg paradox). Consider independent fair coin tosses and let  $T$  be the time of the first Head[cite: 282]:

$$\mathbb{P}(T = k) = 2^{-k}, \quad k = 1, 2, \dots$$

Define the payoff  $X = 2^T$ [cite: 285]. Then  $T < \infty$  almost surely, hence  $X < \infty$  almost surely[cite: 286]. However[cite: 286],

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} 2^k \mathbb{P}(T = k) = \sum_{k=1}^{\infty} 2^k \cdot 2^{-k} = \sum_{k=1}^{\infty} 1 = +\infty$$

## 4 Convergence of integrals

There are several notions of convergence for random variables[cite: 289]. For questions about interchanging limits and expectations/integrals, the most important notion is almost sure convergence together with additional conditions[cite: 289].

**Definition 4.1** (Almost sure convergence). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $X_1, X_2, \dots$  and  $X$  be random variables on it[cite: 290]. We say that  $X_n$  converges to  $X$  almost surely (a.s.), and write  $X_n \rightarrow X$  a.s., if [cite: 291]

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)\right\}\right) = 1$$

Equivalently[cite: 292],

$$\mathbb{P}(\{\omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq X(\omega)\}) = 0$$

A common hope is that if  $X_n \rightarrow X$  a.s., then  $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$ [cite: 294]. This is true under additional assumptions, but fails in general[cite: 295].

**Example 4.1.** Let  $(\Omega, \mathcal{F}, \mathbb{P}) = ([0, 1], \mathcal{B}([0, 1]), \lambda)$ , where  $\lambda$  is the uniform probability measure[cite: 296]. Define [cite: 296]

$$X_n(\omega) = n1_{(0, 1/n)}(\omega)$$

Then for every  $\omega \in (0, 1]$ , we have  $X_n(\omega) = 0$  for all sufficiently large  $n$ [cite: 298]. Since  $\mathbb{P}(\{0\}) = 0$ , it follows that  $X_n \rightarrow 0$  almost surely[cite: 298]. However[cite: 298],

$$\mathbb{E}[X_n] = \int_0^1 n1_{(0, 1/n)}(\omega) d\omega = n \cdot \frac{1}{n} = 1 \text{ for all } n$$

Thus  $\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = 1$  but  $\mathbb{E}[0] = 0$ [cite: 300]. So, in general[cite: 300],

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] \neq \mathbb{E}\left[\lim_{n \rightarrow \infty} X_n\right]$$

### 4.1 Monotone Convergence Theorem

Let  $X_1, X_2, \dots$  be nonnegative random variables on  $(\Omega, \mathcal{F}, \mathbb{P})$  such that [cite: 303]

$$0 \leq X_1(\omega) \leq X_2(\omega) \leq \dots \text{ for all } \omega,$$

and suppose  $X_n(\omega) \rightarrow X(\omega)$  pointwise (hence a.s.)[cite: 305]. Then [cite: 305]

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X] \quad (\text{allowing the value } +\infty)$$

## 4.2 Dominated Convergence Theorem

Let  $X_1, X_2, \dots$  be random variables such that  $X_n \rightarrow X$  almost surely [cite: 309]. Assume there exists an integrable random variable  $Y$  with  $\mathbb{E}[|Y|] < \infty$  such that [cite: 309, 310]

$$|X_n| \leq Y \text{ almost surely for all } n$$

Then  $X$  is integrable and [cite: 311],

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] = \mathbb{E}[X]$$

## 5 Independence

Intuitively, two events are independent if learning that one occurred does not change our assessment of the probability of the other [cite: 315].

**Definition 5.1.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $A, B \in \mathcal{F}$  [cite: 316]. We say that  $A$  and  $B$  are independent if [cite: 316],

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

**Example 5.1** (Two coin tosses). Let  $\Omega = \{HH, HT, TH, TT\}$  with  $\mathbb{P}(H) = p, \mathbb{P}(T) = q = 1 - p$  so that [cite: 319]:

$$\mathbb{P}(HH) = p^2, \quad \mathbb{P}(HT) = pq, \quad \mathbb{P}(TH) = pq, \quad \mathbb{P}(TT) = q^2$$

Let  $A = \{\text{Head on the first toss}\} = \{HH, HT\}$ ,  $B = \{\text{Head on the second toss}\} = \{HH, TH\}$  [cite: 320, 322]. Then  $\mathbb{P}(A) = p, \mathbb{P}(B) = p$ , and [cite: 322]

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = p^2 = \mathbb{P}(A)\mathbb{P}(B)$$

so  $A$  and  $B$  are independent [cite: 323].

**Definition 5.2** (Independence of  $\sigma$ -algebras). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $\mathcal{G}, \mathcal{H}$  be sub- $\sigma$ -algebras of  $\mathcal{F}$  [cite: 325]. We say that  $\mathcal{G}$  and  $\mathcal{H}$  are independent if [cite: 325]

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

for all  $A \in \mathcal{G}, B \in \mathcal{H}$  [cite: 327].

**Definition 5.3** (Independence of random variables). Let  $X$  and  $Y$  be random variables on  $(\Omega, \mathcal{F}, \mathbb{P})$  [cite: 328]. We say that  $X$  and  $Y$  are independent if the  $\sigma$ -algebras they generate are independent [cite: 328, 329]:  $\sigma(X)$  and  $\sigma(Y)$  are independent [cite: 330]. More generally, we say that  $X$  is independent of a sub- $\sigma$ -algebra  $\mathcal{G}$  if  $\sigma(X)$  and  $\mathcal{G}$  are independent [cite: 331].

*Remark* (Event form of independence of random variables). Recall that  $\sigma(X)$  is generated by events of the form  $\{X \in C\}$  where  $C$  ranges over Borel subsets of  $\mathbb{R}$  [cite: 332]. Thus  $X$  and  $Y$  are independent if and only if [cite: 332]

$$\mathbb{P}(X \in C, Y \in D) = \mathbb{P}(X \in C)\mathbb{P}(Y \in D) \text{ for all Borel } C, D \subseteq \mathbb{R}$$

**Example 5.2.** Toss a coin twice with  $\mathbb{P}(H) = p \in (0, 1)$  and  $\mathbb{P}(T) = q = 1 - p$ . Let  $\Omega = \{HH, HT, TH, TT\}$  with the product probabilities [cite: 334]. Define two random variables [cite: 335],

$$X(\omega) = 1_{\{\text{first toss is } H\}}$$

$$Y(\omega) = 1_{\{\text{second toss is } H\}}$$

Thus  $X, Y \in \{0, 1\}$ [cite: 338]. Since  $X$  takes only the values 0 and 1[cite: 339],

$$\sigma(X) = \{\emptyset, \Omega, \{X = 1\}, \{X = 0\}\}$$

Here,  $\{X = 1\} = \{HH, HT\}$ ,  $\{X = 0\} = \{TH, TT\}$ [cite: 340, 344]. Similarly[cite: 341],

$$\sigma(Y) = \{\emptyset, \Omega, \{Y = 1\}, \{Y = 0\}\}$$

with  $\{Y = 1\} = \{HH, TH\}$ ,  $\{Y = 0\} = \{HT, TT\}$ [cite: 346, 347].

To prove that  $X$  and  $Y$  are independent, we must show that  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$  for all  $A \in \sigma(X)$ ,  $B \in \sigma(Y)$ [cite: 348, 349]. Because  $\sigma(X)$  and  $\sigma(Y)$  are generated by  $\{X = 1\}$  and  $\{Y = 1\}$  (and complements), it is enough to verify the four basic cases where  $A \in \{\{X = 1\}, \{X = 0\}\}$ ,  $B \in \{\{Y = 1\}, \{Y = 0\}\}$ [cite: 350, 353]:

$$\mathbb{P}(\{X = 1\} \cap \{Y = 1\}) = \mathbb{P}(\{HH\}) = p^2 = \mathbb{P}(X = 1)\mathbb{P}(Y = 1) = p^2$$

$$\mathbb{P}(\{X = 1\} \cap \{Y = 0\}) = \mathbb{P}(\{HT\}) = pq = \mathbb{P}(X = 1)\mathbb{P}(Y = 0) = pq$$

$$\mathbb{P}(\{X = 0\} \cap \{Y = 1\}) = \mathbb{P}(\{TH\}) = qp = \mathbb{P}(X = 0)\mathbb{P}(Y = 1) = qp$$

$$\mathbb{P}(\{X = 0\} \cap \{Y = 0\}) = \mathbb{P}(\{TT\}) = q^2 = \mathbb{P}(X = 0)\mathbb{P}(Y = 0) = q^2$$

Hence  $\sigma(X)$  and  $\sigma(Y)$  are independent, so  $X$  and  $Y$  are independent random variables[cite: 356].

**Definition 5.4.** Let  $\mathcal{G}_1, \dots, \mathcal{G}_n$  be sub- $\sigma$ -algebras of  $\mathcal{F}$ [cite: 357]. We say that they are independent if [cite: 357]

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \dots \mathbb{P}(A_n) \text{ for all } A_i \in \mathcal{G}_i$$

**Definition 5.5.** A sequence of sub- $\sigma$ -algebras  $\mathcal{G}_1, \mathcal{G}_2, \dots$  is called independent if for every  $n$ , the family  $\mathcal{G}_1, \dots, \mathcal{G}_n$  is independent[cite: 359]. A sequence of random variables  $X_1, X_2, \dots$  is called independent if the sequence of  $\sigma$ -algebras  $\sigma(X_1), \sigma(X_2), \dots$  is independent[cite: 360].

**Example 5.3.** On the infinite coin-toss space, let  $Y_k(\omega) = 1_{\{\omega_k = H\}}$ [cite: 361, 362]. Then the  $\sigma$ -algebras  $\sigma(Y_1), \sigma(Y_2), \dots$  (equivalently, the information from each toss) are independent, and the random variables  $Y_1, Y_2, \dots$  are independent[cite: 363].

Let  $X$  and  $Y$  be independent random variables, and let  $f$  and  $g$  be Borel-measurable functions on  $\mathbb{R}$ [cite: 364]. Then  $f(X)$  and  $g(Y)$  are independent[cite: 364]. Essentially, if  $X$  and  $Y$  are independent random variables then  $f(X)$  and  $g(Y)$  are also independent[cite: 365].

**Definition 5.6.** Let  $X$  and  $Y$  be random variables on  $(\Omega, \mathcal{F}, \mathbb{P})$ [cite: 367]. The joint distribution measure  $\mu_{X,Y}$  on  $\mathbb{R}^2$  is defined by[cite: 368]:

$$\mu_{X,Y}(C) = \mathbb{P}((X, Y) \in C), \quad C \in \mathcal{B}(\mathbb{R}^2)$$

**Definition 5.7.** The joint cumulative distribution function of  $(X, Y)$  is [cite: 371]

$$F_{X,Y}(a, b) = \mu_{X,Y}((-\infty, a] \times (-\infty, b]) = \mathbb{P}(X \leq a, Y \leq b), \quad a, b \in \mathbb{R}$$

**Definition 5.8.** A nonnegative Borel-measurable function  $f_{X,Y}(x, y)$  is called a joint density of  $(X, Y)$  if [cite: 374]

$$\mu_{X,Y}(C) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} 1_C(x, y) f_{X,Y}(x, y) dy dx, \quad C \in \mathcal{B}(\mathbb{R}^2)$$

Equivalently[cite: 375],

$$F_{X,Y}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx, \quad a, b \in \mathbb{R}$$

**Definition 5.9.** The marginal distribution measures of  $X$  and  $Y$  are[cite: 379],

$$\mu_X(A) = \mu_{X,Y}(A \times \mathbb{R}) = \mathbb{P}(X \in A)$$

$$\mu_Y(B) = \mu_{X,Y}(\mathbb{R} \times B) = \mathbb{P}(Y \in B)$$

for Borel sets  $A, B \subseteq \mathbb{R}$ [cite: 382]. The marginal CDFs are  $F_X(a) = \mathbb{P}(X \leq a)$  and  $F_Y(b) = \mathbb{P}(Y \leq b)$ [cite: 382]. If a joint density  $f_{X,Y}$  exists, the marginal densities (when they exist) are given by [cite: 382]

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Let  $X$  and  $Y$  be random variables[cite: 385]. The following are equivalent[cite: 385]:

1.  $X$  and  $Y$  are independent[cite: 386].
2. The joint distribution measure factors:  $\mu_{X,Y}(A \times B) = \mu_X(A)\mu_Y(B)$  for all Borel  $A, B \subseteq \mathbb{R}$ [cite: 387, 388, 390].
3. The joint CDF factors:  $F_{X,Y}(a, b) = F_X(a)F_Y(b)$  for all  $a, b \in \mathbb{R}$ [cite: 389, 391, 392].

## 6 Conditional expectation and martingales

**Definition 6.1.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space, let  $\mathcal{G}$  be a sub- $\sigma$  algebra of  $\mathcal{F}$ , and let  $X$  be an integrable random variable[cite: 394]. The conditional expectation of  $X$  given  $\mathcal{G}$ , denoted as  $\mathbb{E}[X|\mathcal{G}]$ , is a random variable that satisfies[cite: 395]:

1.  $\mathbb{E}[X|\mathcal{G}]$  is  $\mathcal{G}$ -measurable [cite: 396]
2.  $\int_A \mathbb{E}[X|\mathcal{G}](\omega) d\mathbb{P}(\omega) = \int_A X(\omega) d\mathbb{P}(\omega), \quad \forall A \in \mathcal{G}$  [cite: 397]

*Remark.* Suppose  $X$  and  $Y$  are two random variables, sometimes we write  $\mathbb{E}[X|Y]$  as a shorthand for  $\mathbb{E}[X|\sigma(Y)]$ [cite: 398].

Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space and let  $\mathcal{G}$  be a sub- $\sigma$  algebra of  $\mathcal{F}$ [cite: 399].

1. If  $X, Y$  are integrable random variables and  $c_1, c_2$  are constants, then[cite: 401]:

$$\mathbb{E}[c_1 X + c_2 Y|\mathcal{G}] = c_1 \mathbb{E}[X|\mathcal{G}] + c_2 \mathbb{E}[Y|\mathcal{G}]$$

2. If  $X, Y, XY$  are integrable random variables and  $X$  is  $\mathcal{G}$ -measurable[cite: 403]:

$$\mathbb{E}[XY|\mathcal{G}] = X \mathbb{E}[Y|\mathcal{G}]$$

3. If  $\mathcal{H}$  is a sub- $\sigma$  algebra of  $\mathcal{G}$  and  $X$  is an integrable random variable, then[cite: 405]:

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X|\mathcal{H}]$$

4. If  $X$  is integrable and independent of  $\mathcal{G}$ , then[cite: 407]:

$$\mathbb{E}[X|\mathcal{G}] = \mathbb{E}[X]$$

**Example 6.1.** Let  $\Omega = \{1, 2, 3, 4\}$  with  $\mathbb{P}(\{i\}) = \frac{1}{4}$  and define  $X(\omega) = \omega$ [cite: 409, 410]. Let [cite: 410]

$$\mathcal{G} = \sigma(\{1, 2\}, \{3, 4\}) = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$$

By definition,  $\mathbb{E}[X|\mathcal{G}]$  is a  $\mathcal{G}$ -measurable random variable  $Y$  satisfying  $\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}$ ,  $\forall A \in \mathcal{G}$ [cite: 412, 413, 414]. Since  $\mathcal{G}$  has elements  $\{1, 2\}$  and  $\{3, 4\}$ , any  $\mathcal{G}$ -measurable  $Y$  must be constant on each of these sets[cite: 415]. Write  $Y(1) = Y(2) = a$ ,  $Y(3) = Y(4) = b$ [cite: 415, 418].

For  $A = \{1, 2\}$ [cite: 416, 417]:

$$\int_{1,2} Y d\mathbb{P} = a \cdot \frac{1}{4} + a \cdot \frac{1}{4} = \frac{a}{2}, \quad \int_{1,2} X d\mathbb{P} = 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{4} = \frac{3}{4},$$

hence  $\frac{a}{2} = \frac{3}{4}$  and  $a = \frac{3}{2}$ [cite: 420].

For  $A = \{3, 4\}$ [cite: 421, 422]:

$$\int_{3,4} Y d\mathbb{P} = b \cdot \frac{1}{4} + b \cdot \frac{1}{4} = \frac{b}{2}, \quad \int_{3,4} X d\mathbb{P} = 3 \cdot \frac{1}{4} + 4 \cdot \frac{1}{4} = \frac{7}{4},$$

so  $\frac{b}{2} = \frac{7}{4}$  and  $b = \frac{7}{2}$ [cite: 424].

Therefore[cite: 425],

$$\mathbb{E}[X|\mathcal{G}](\omega) = \begin{cases} \frac{3}{2}, & \omega \in \{1, 2\}, \\ \frac{7}{2}, & \omega \in \{3, 4\}. \end{cases}$$

**Example 6.2** (Two coin tosses: conditioning on the first toss). Let  $\Omega = \{HH, HT, TH, TT\}$  with each outcome having probability  $\frac{1}{4}$ [cite: 427, 428]. Define  $X = 1_{\{\text{second toss is Head}\}}$ , so that  $X(HH) = X(TH) = 1$  and  $X(HT) = X(TT) = 0$ [cite: 429, 430]. Let [cite: 430]

$$\mathcal{G} = \sigma(\{\text{first toss is Head}\}) = \sigma(\{HH, HT\}) = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}$$

By definition,  $\mathbb{E}[X|\mathcal{G}]$  is a  $\mathcal{G}$ -measurable random variable  $Y$  satisfying  $\int_A Y d\mathbb{P} = \int_A X d\mathbb{P}$ ,  $\forall A \in \mathcal{G}$ [cite: 433, 434]. Since  $\mathcal{G}$  has atoms  $\{HH, HT\}$  and  $\{TH, TT\}$ , any  $\mathcal{G}$ -measurable  $Y$  must be constant on each of these sets[cite: 435]. Write  $Y(HH) = Y(HT) = a$ ,  $Y(TH) = Y(TT) = b$ [cite: 436, 439].

We determine  $a$  and  $b$  using the defining identity[cite: 437]. For  $A = \{HH, HT\}$ [cite: 437]:

$$\int_{\{HH, HT\}} Y d\mathbb{P} = a \cdot \frac{1}{4} + a \cdot \frac{1}{4} = \frac{a}{2}$$

$$\int_{\{HH, HT\}} X d\mathbb{P} = X(HH) \frac{1}{4} + X(HT) \frac{1}{4} = 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = \frac{1}{4}$$

Thus  $\frac{a}{2} = \frac{1}{4}$  so  $a = \frac{1}{2}$ [cite: 441].

For  $A = \{TH, TT\}$ ,  $\int_{\{TH, TT\}} Y d\mathbb{P} = b \cdot \frac{1}{4} + b \cdot \frac{1}{4} = \frac{b}{2}$ , and[cite: 443]:

$$\int_{\{TH, TT\}} X d\mathbb{P} = X(TH) \frac{1}{4} + X(TT) \frac{1}{4} = 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = \frac{1}{4}$$

Hence  $\frac{b}{2} = \frac{1}{4}$ , giving  $b = \frac{1}{2}$ [cite: 445].

Therefore[cite: 446],

$$\mathbb{E}[X|\mathcal{G}](\omega) = \begin{cases} \frac{1}{2}, & \omega \in \{HH, HT\}, \\ \frac{1}{2}, & \omega \in \{TH, TT\}, \end{cases}$$

so

$$\mathbb{E}[X|\mathcal{G}] = \frac{1}{2} \quad \text{a.s.}$$

This reflects the fact that, for independent tosses, learning the first toss provides no information about the second[cite: 451].

**Definition 6.2.** Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space, let  $T$  be a fixed number, and let  $\mathcal{F}(t), 0 \leq t \leq T$ , be a filtration of sub- $\sigma$ -algebras of  $\mathcal{F}$ [cite: 452]. Consider an adapted stochastic process  $M(t), 0 \leq t \leq T$ [cite: 452]:

1. If  $\mathbb{E}[M(t)|\mathcal{F}(s)] = M(s)$  for all  $0 < s < t < T$ , then  $M(t)$  is said to be a **martingale**[cite: 453].
2. If  $\mathbb{E}[M(t)|\mathcal{F}(s)] \geq M(s)$  for all  $0 \leq s \leq t \leq T$  then  $M(t)$  is said to be a **submartingale**[cite: 454].
3. If  $\mathbb{E}[M(t)|\mathcal{F}(s)] \leq M(s)$  for all  $0 \leq s \leq t \leq T$ , then  $M(t)$  is said to be a **supermartingale**[cite: 455].